Notes: Kogan, Livdan, and Yaron (JF, 2009)

Oil Futures Prices in a Production Economy with Investment Constraints

Contents

1 Big picture

- **Question.** What economic mechanism explains the conditional volatility of oil futures prices, especially the fact that futures-price volatility is high both when the forward curve is steeply backwardated and when it is steeply in contango?
- **Setting.** The paper studies NYMEX light sweet crude oil futures from **1983 to 2000**, focusing on contracts with up to **12 months** to delivery.
- **Main empirical fact.** The relationship between futures-price volatility and the slope of the forward curve is **nonmonotone and V-shaped**. Volatility rises when the curve is very negatively sloped and also when the curve is very positively sloped.
- **Why this is important.** Standard storage models emphasize inventories and tend to predict a monotone relationship between volatility and backwardation. The V-shape points toward a broader production-side mechanism.
- **Main theoretical idea.** Oil supply is not perfectly flexible. Production capacity is built through investment, and investment is constrained by irreversibility and an upper bound on how fast firms can expand capacity.
- **Economic mechanism.** When capacity is far above demand, firms stop investing because investment is irreversible. When capacity is far below demand, firms invest as fast as possible but cannot expand instantly. In both cases, supply is relatively inelastic and demand shocks mainly move prices.
- **Connection to the forward curve.** The slope of the forward curve is large in absolute value when the current spot price is far from its long-run level. These are exactly the states in which investment constraints bind and futures-price volatility is high.
- **Model.** The authors build a competitive continuous-time production economy with a demand shock, capital accumulation, irreversible investment, and an upper bound on investment proportional to aggregate capital.
- **Futures pricing.** Futures prices are risk-neutral expectations of future spot prices. The spot price is endogenous because it clears the market between stochastic demand and constrained production capacity.
- **Estimation.** The model is estimated by simulated method of moments. The moments include unconditional futures-price volatilities, moments of the forward-curve slope, expected oil-consumption growth, and the V-shape regression coefficients.
- **Main quantitative result.** The estimated model reproduces a slowly declining term structure of futures-price volatility and a V-shaped relation between volatility and the forward-curve slope.
- **Economic takeaway.** Futures-price volatility is not only an inventory phenomenon. Time-varying supply elasticity caused by production and investment constraints is a central driver of commodity futures dynamics.

2 Data and measurement (oil futures facts and key objects)

2.1 Why crude oil futures are the right laboratory

- Crude oil is the basic upstream input for other petroleum products, so production capacity and investment constraints are especially relevant.
- The NYMEX crude oil futures market has high liquidity over short and medium maturities.
- Oil prices display classic commodity features: backwardation, mean reversion, heteroscedasticity, and volatility related to the shape of the forward curve.
- The production side is economically important because oil wells, rigs, extraction capacity, and delivery infrastructure cannot adjust instantly.

Interpretation. Crude oil is useful because it is both financially traded and physically produced under slow capacity adjustment. The paper can therefore connect futures prices to real investment frictions.

2.2 Futures-price data

- The data are daily NYMEX light sweet crude oil futures prices.
- The sample period is **1983–2000**.
- Contracts are sorted by delivery horizon: the "1-month" contract has the nearest delivery date, the "2-month" contract has the next-nearest delivery date, and so on.
- The analysis uses maturities up to **12 months**, where liquidity and data availability are good.
- Depending on maturity, the sample contains roughly **2,500–3,500 daily observations**.

Interpretation. The data give a daily term structure of oil futures prices. This allows the authors to measure both the slope of the forward curve and the volatility of price changes at different maturities.

2.3 Daily futures-price changes

The paper works with daily percentage price changes rather than levels:

$$R(t,T) = \frac{P(t,T)}{P(t-1,T)} - 1, \tag{1}$$

where $P(t,T)$ is the futures price on day t for a contract with time to delivery T.

Interpretation. Price levels can contain trends and seasonalities. Percentage changes are more appropriate for studying volatility.

2.4 Forward-curve slope

The key slope variable is the log ratio of the 6-month and 3-month futures prices:

$$SL(t) = \log\left(\frac{P(t,6)}{P(t,3)}\right). \tag{2}$$

The regressions use the demeaned slope

$$\widetilde{SL}(t) = SL(t) - E[SL(t)]. \tag{3}$$

Interpretation. A negative slope corresponds to backwardation: near-term oil is priced above longer-maturity oil. A positive slope corresponds to contango: longer-maturity oil is priced above near-term oil. Demeaning focuses on deviations from the average slope.

2.5 Conditional volatility object

The paper uses $|R(t,T)|$ as a simple daily measure of realized volatility for maturity T.

Interpretation. Absolute daily returns are noisy, but they provide a transparent way to test whether high or low basis states are associated with high volatility.

2.6 Initial unconditional and conditional volatility facts

- Unconditionally, futures-price volatility declines with maturity, consistent with the Samuelson effect.
- Volatility is higher when the forward curve is backwardated than when it is in contango.
- This conditional fact is often interpreted through storage theories: low inventories and high convenience yields are associated with backwardation and high spot-price volatility.

Interpretation. The standard split between backwardation and contango is useful but incomplete. The paper shows that a simple binary split hides a sharper nonlinear pattern.

2.7 Main empirical object: the V-shape

- The central object is the conditional relation between $|R(t,T)|$ and $\widetilde{SL}(t-1)$.
- The paper finds that volatility is high when the demeaned slope is very negative and also when it is very positive.
- Therefore, volatility is not simply decreasing or increasing in the basis; it is approximately V-shaped.

Interpretation. The slope of the forward curve is a state variable. Extreme slope states reveal periods when the market expects the current imbalance between demand and capacity to unwind over time.

2.8 Key structural objects in the model

- K_t: productive capital or capacity in the oil industry.
- Q_t: oil output.
- Y_t: demand-shock level.
- S_t: spot price.
- I_t: investment rate.
- $\bar{\imath}$: maximum feasible investment rate as a fraction of aggregate capital.
- δ: depreciation rate of productive capital.
- γ: curvature parameter in the inverse demand curve; $1/\gamma$ is the absolute price elasticity of demand in the model.

Interpretation. The state is not inventory. The central state is the relation between demand and productive capacity.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Linear volatility-basis regression

The first empirical specification is

$$|R(t,T)| = \alpha_T + \beta_T \widetilde{SL}(t-1) + \varepsilon_T(t). \tag{4}$$

Interpretation. This regression asks whether volatility rises or falls with the slope of the forward curve. A single β_T imposes monotonicity, so it cannot capture a V-shape.

Result. The estimated β_T is negative for the reported maturities. For example, Table I reports $\beta_T = -0.0743$ at 1 month, -0.0462 at 5 months, and -0.0419 at 10 months.

3.2 Piecewise-linear V-shape regression

To allow different slopes in backwardation-like and contango-like states, the paper estimates

$$|R(t,T)| = \alpha_T + \beta_{1,T}\big(\widetilde{SL}(t-1)\big)^+ + \beta_{2,T}\big(\widetilde{SL}(t-1)\big)^- + \varepsilon_T(t), \tag{5}$$

where $(X)^+$ denotes the positive part of X and $(X)^-$ denotes the negative part of X.

Interpretation. A V-shape means $\beta_{1,T} > 0$ and $\beta_{2,T} < 0$. When the slope is positive, a larger slope raises volatility. When the slope is negative, a more negative slope also raises volatility because $\beta_{2,T} < 0$.

Result. The estimates have the predicted signs. Table I reports:

- 1 month: $\beta_{1,T} = 0.3191$ and $\beta_{2,T} = -0.3505$.
- 5 months: $\beta_{1,T} = 0.1791$ and $\beta_{2,T} = -0.2039$.
- 10 months: $\beta_{1,T} = 0.1308$ and $\beta_{2,T} = -0.1632$.

Interpretation. The data reject the view that volatility is just a monotone function of backwardation. Both extreme contango and extreme backwardation are high-volatility states.

3.3 Nonparametric volatility-basis relation

The paper also estimates the relation nonparametrically using locally weighted methods.

Interpretation. This checks that the V-shape is not mechanically imposed by the piecewise-linear regression. The nonparametric curves for 2-, 4-, 6-, and 8-month contracts show a clear nonmonotone V-shape.

3.4 GARCH robustness regressions

The authors check whether the V-shape survives after controlling for conventional volatility dynamics. One specification is

$$|R(t,T)| = \alpha_T + \beta_{G,T}\sigma^{GARCH}(t,T) + \beta_{1,T}\big(\widetilde{SL}(t-1)\big)^+ + \beta_{2,T}\big(\widetilde{SL}(t-1)\big)^- + \varepsilon_T(t). \tag{6}$$

A second specification uses a rolling GARCH prediction:

$$\widehat{\sigma}^{GARCH}(t,T) = \alpha_T + \beta_{G,T}\widehat{\sigma}^{GARCH}(t-1,T) + \beta_{1,T}\big(\widetilde{SL}(t-1)\big)^+ + \beta_{2,T}\big(\widetilde{SL}(t-1)\big)^- + \varepsilon_T(t). \tag{7}$$

Interpretation. These regressions ask whether the slope still matters after accounting for predictable time-series volatility. The V-shape coefficients remain statistically significant for most reported maturities.

3.5 Production technology

The model is a continuous-time infinite-horizon competitive production economy. Firms produce a nonstorable good using capital:

$$Q_t = XK_t. \tag{8}$$

The authors normalize $X = 1$.

Interpretation. Output is proportional to capacity. The important friction is not a complicated production function; it is the inability to adjust capacity freely.

3.6 Capital accumulation

Capital evolves according to

$$dK_t = (I_t - \delta K_t)dt, \tag{9}$$

where I_t is investment and δ is depreciation.

Interpretation. Capacity increases through investment and falls through depreciation. Because capacity is durable, current investment decisions affect future supply.

3.7 Investment constraints

Investment is irreversible and its rate is bounded:

$$I_t \in [0, \bar{\imath} K_t^A], \tag{10}$$

where K_t^A is aggregate capital.

Interpretation. Firms cannot disinvest, and they cannot expand capacity infinitely fast. These two constraints create low supply elasticity at both extremes: too much capacity and too little capacity.

3.8 Firm value

Firms sell output at the spot price S_t. Under complete markets, firm value is

$$V_0 = E_0 \left[\int_0^\infty e^{-rt}(S_t Q_t - I_t)dt \right], \tag{11}$$

where r is the risk-free rate.

Interpretation. Firms invest when the present value of the extra output generated by an additional unit of capacity exceeds the unit cost of investment.

3.9 Demand curve and demand shocks

The representative demand curve is

$$Q_t = Y_t S_t^{-1/\gamma}, \qquad Q_t \in (0, \infty), \tag{12}$$

where unexpected changes in Y_t are demand shocks. Under the risk-neutral measure,

$$\frac{dY_t}{Y_t} = \mu_y dt + \sigma_y dW_t. \tag{13}$$

Interpretation. A positive demand shock raises demand at any given spot price. The parameter $1/\gamma$ is the absolute price elasticity of demand.

3.10 Competitive equilibrium and investment trigger

In equilibrium, the aggregate capital stock evolves as

$$dK_t = (I_t^A - \delta K_t)dt. \tag{14}$$

The optimal investment policy has a threshold form:

$$I_t^* = \begin{cases} \bar{\imath} K_t^A, & S_t \geq S^*, \\ 0, & S_t < S^*. \end{cases} \tag{15}$$

Interpretation. When the spot price is high, capacity is valuable and firms invest at the maximum rate. When the spot price is low, capacity is not valuable enough to justify new investment, so investment stops.

3.11 Risk-neutral spot-price dynamics

The equilibrium spot-price process is

$$\frac{dS_t}{S_t} = \left[-\gamma \left(\bar{\imath} \mathbf{1}_{[S_t \geq S^*]} - \mu^- \right) + \frac{\gamma^2 \sigma_y^2}{2} \right] dt + \gamma \sigma_y dW_t, \tag{16}$$

where

$$\mu^- = \delta + \mu_y - \frac{1}{2}\sigma_y^2, \qquad \mu^+ = \bar{\imath} - \mu^-. \tag{17}$$

Interpretation. The drift of the spot price changes at the investment trigger. Above the trigger, maximum investment gradually expands capacity and pushes prices back down. Below the trigger, zero investment and depreciation gradually reduce excess capacity and push prices back up.

3.12 Stationary distribution

The spot-price process has a stationary long-run distribution when the parameter restrictions are satisfied. The density is piecewise and centered around the investment trigger S^*.

Interpretation. The economy does not wander permanently away from the trigger. When prices are high, capacity expansion pushes them down. When prices are low, depreciation and no investment push them up.

3.13 Futures prices

A futures price is the risk-neutral expectation of the future spot price:

$$P(t, T) = E_t[S_{t+T}], \qquad T > 0. \tag{18}$$

Interpretation. Futures prices reflect the expected future path of the spot price under constrained capacity adjustment. Longer-maturity futures are less sensitive to the current spot price than short-maturity futures.

3.14 Why the model predicts a V-shape

- If S_t is far below S^*, investment is zero. Supply cannot fall instantly because capital only depreciates gradually. Demand shocks therefore move prices strongly.
- If S_t is far above S^*, investment is at its upper bound. Supply cannot rise instantly because investment capacity is limited. Demand shocks again move prices strongly.
- If S_t is near S^*, small demand shocks can be absorbed more flexibly through changes in investment. Price volatility is lower.

Interpretation. The forward-curve slope is large in absolute value when S_t is far from its long-run level. Therefore, high absolute slope states correspond to high volatility states.

3.15 Simulated method of moments

The paper estimates structural parameters by minimizing

$$J_T = G_N(X,\theta)' W_T G_N(X,\theta), \tag{19}$$

where

$$G_N(X,\theta) = m_N(\theta) - F_T(X). \tag{20}$$

Here $F_T(X)$ is a vector of empirical moments from the data, and $m_N(\theta)$ is the average of the same moments computed from simulated model data.

Interpretation. The model is too nonlinear to estimate by closed-form likelihood. SMM chooses parameters so that simulated futures prices match the empirical moments.

3.16 Moments used in estimation

The estimation targets include:

- average daily return on a fully collateralized 3-month futures position;
- unconditional volatility of daily price changes at several futures maturities;
- mean, volatility, and 30-day autocorrelation of the forward-curve slope;
- expected growth rate of oil consumption;
- V-shape coefficients $\beta_{1,T}$ and $\beta_{2,T}$ for maturities of 3, 6, and 9 months.

Interpretation. The estimation is designed to fit both standard unconditional futures-price facts and the paper's main conditional volatility fact.

3.17 Oil-consumption growth moment

The model-implied expected growth rate of oil consumption is

$$g_c = \bar{\imath}\,\mathrm{Pr}(S > S^*) - \delta = \lambda + \mu_y - \frac{1}{2}\sigma_y^2, \tag{21}$$

where λ is the market price of oil-price risk in the paper's notation.

Interpretation. Futures-price data alone cannot separately identify all structural parameters. Oil-consumption growth provides an additional real-side moment.

4 Identification and model credibility (what makes the exercise informative)

4.1 What identifies the empirical V-shape

- The V-shape is identified from daily variation in the lagged slope of the forward curve and the absolute returns of futures contracts.
- The key test is whether $\beta_{1,T} > 0$ and $\beta_{2,T} < 0$ in the piecewise regression.
- The pattern is visible in parametric regressions and in nonparametric curves.

Interpretation. The empirical contribution is not just that backwardation is volatile. It is that volatility is low near the middle of the slope distribution and high at both tails.

4.2 Why storage-only models struggle

- Traditional storage models explain high volatility in backwardation through low inventories and high convenience yields.
- They generally imply a monotone relation between volatility and the slope of the forward curve.
- They also have difficulty producing the slow decay of futures-price volatility across maturities.

Interpretation. Storage is important for the short end of the curve, but the V-shape at medium maturities suggests that production capacity constraints also matter.

4.3 Why production constraints matter

- Production capacity cannot be scrapped immediately when demand is low.
- Production capacity cannot be expanded immediately when demand is high.
- Both constraints reduce supply elasticity in extreme states.
- Reduced supply elasticity turns demand shocks into larger price movements.

Interpretation. The model's core variable is time-varying supply elasticity. The sign of the forward-curve slope matters less than its absolute size.

4.4 What identifies structural parameters

- The term structure of unconditional volatility helps identify the magnitude and persistence of spot-price dynamics.
- The mean, volatility, and autocorrelation of the slope help identify how fast the economy moves around the investment trigger.
- The V-shape coefficients identify how volatility changes as the economy moves away from the trigger.
- Oil-consumption growth helps discipline the real side of the model.
- The authors fix $r = 0.02$ and $\delta = 0.12$ because the futures-price moments alone do not independently identify all structural parameters.

Interpretation. The exercise is disciplined but not fully nonparametric. The model must use outside restrictions because futures prices alone cannot pin down every deep parameter separately.

4.5 Parameter estimates and interpretation

- The estimated demand curvature is $\gamma = 3.1484$, implying a demand price elasticity around -0.32.
- The maximum investment rate is $\bar{\imath} = 0.2362$ per year.
- The demand-shock drift is $\mu_y = 0.01438$.
- The demand-shock volatility is $\sigma_y = 0.1043$.
- The fixed depreciation rate is $\delta = 0.1200$.
- The risk premium parameter is small, $\lambda = 2.47 \times 10^{-4}$.

Interpretation. The estimates imply inelastic oil demand, slow but meaningful capacity expansion, and spot-price volatility close to observed short-maturity futures volatility.

4.6 What the model matches well

- The model matches the declining but slow-moving term structure of futures-price volatility.
- The model matches the sign pattern of the V-shape coefficients.
- The model matches the volatility of the forward-curve slope almost exactly: the model value is about 0.0285 versus 0.0287 in the data.
- The model gives a 30-day slope autocorrelation of about 0.82 compared with 0.77 in the data.

- The model-implied expected growth rate of oil consumption is about 0.9 percent, close to the world growth rate of about 1.25 percent.

Interpretation. A simple one-factor production model can explain several important futures-price facts that are difficult for a simple storage model.

4.7 What the model does not fully match

- The model does not include storage, so it is less suited to explaining the very short end of the forward curve.
- The fit of the V-shape coefficients is weaker at 1- and 2-month maturities.
- The model assumes a constant risk premium, which is a simplifying restriction.
- The model abstracts from exhaustibility of oil reserves, seasonality, geopolitical shocks, and strategic behavior.

Interpretation. The paper is not claiming that storage is irrelevant. Rather, it shows that production constraints are needed to explain an important conditional volatility pattern.

4.8 Role of risk premia

- The model is solved under the risk-neutral measure.
- To simulate under the physical measure, the paper assumes a constant market price of oil-price risk.
- This affects the drift of prices but not the core production-side volatility mechanism.

Interpretation. The V-shape is not mainly a risk-premium story. It comes from the way investment constraints change supply elasticity across states.

4.9 Sensitivity analysis

- Higher demand volatility σ_y raises futures-price volatility across maturities.
- Higher demand curvature γ also raises price volatility because demand is less elastic.
- A higher maximum investment rate $\bar{\imath}$ lets capacity adjust faster, reducing longer-maturity futures volatility and steepening the decline of the volatility term structure.
- Changes in the demand drift μ_y and risk premium λ mostly affect growth-related moments rather than the volatility level.

Interpretation. The model's volatility predictions are driven mainly by demand volatility, demand elasticity, and the speed of investment adjustment.

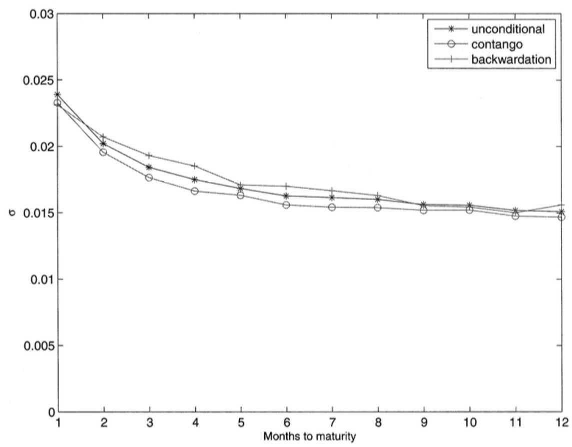

Figure 1. Term structure of volatility, crude oil futures. The data are daily percentage price changes on NYMEX crude oil (CL) futures from 1983 to 2000. σ denotes the standard deviation of daily percentage price changes. The time to maturity is defined as the number of months left until the delivery month. The unconditional standard deviation is constructed using the sample's first and second moments, while standard deviations conditional on backwardation and contango are conditioned on the shape of the forward curve one day prior.

5 Tables and figures

5.1 Figure 1: Term structure of futures-price volatility

Read:

- The figure plots unconditional volatility and volatility conditional on contango or backwardation.
- Unconditional volatility declines with maturity.
- Volatility is generally higher when the forward curve is backwardated.
- This reproduces the familiar commodity-market fact that backwardation is associated with high volatility.

Economic message. The standard backwardation-versus-contango split suggests a storage interpretation, but it does not reveal the full nonlinear pattern.

5.2 Figure 2 and Table I: Conditional volatility regressions

Read:

- The linear regression coefficients β_T are negative at 1, 5, and 10 months.
- The piecewise coefficients have opposite signs: $\beta_{1,T} > 0$ and $\beta_{2,T} < 0$.
- At 1 month, the piecewise coefficients are 0.3191 and -0.3505.
- At 5 months, they are 0.1791 and -0.2039.
- At 10 months, they are 0.1308 and -0.1632.
- The piecewise regressions have much higher R^2 than the linear regressions.

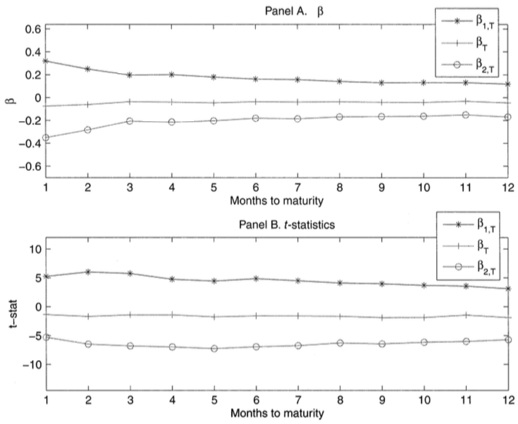

Figure 2. Conditional volatility, crude oil futures. The data are daily percentage price changes on NYMEX crude oil (CL) futures from 1983 to 2000, denoted by $R(t, T)$. Two different specifications are used to relate the instantaneous volatility of daily percentage price changes to the beginning-of-period slope of the forward curve defined as

$$SL(t) = \ln\left[\frac{P(t-1,6)}{P(t-1,3)}\right].$$

Table I
Conditional Volatility, Crude Oil Futures

This table reports results for two different regressions. The data are daily percentage price changes on NYMEX crude oil (CL) futures from 1983 to 2000. The specifications in both panels are the same as in Figure 2. All results are reported for times to maturity equal to 1, 5 and 10 months. All t-statistics are White-adjusted for conditional heteroscedasticity.

	1 Month	5 Months	10 Months
	$\lvert R(t,T)\rvert = \alpha_T + \beta_T \widetilde{SL}(t-1) + \varepsilon_T(t)$		
β_T	−0.0743	−0.0462	−0.0419
(t-stat)	(−1.37)	(−1.77)	(−1.89)
R^2	0.0122	0.0111	0.0117
	$\lvert R(t,T)\rvert = \alpha_T + \beta_{1,T}(\widetilde{SL}(t-1))^+ + \beta_{2,T}(\widetilde{SL}(t-1))^- + \varepsilon_T(t)$		
$\beta_{1,T}$	0.3191	0.1791	0.1308
(t-stat)	(5.25)	(4.42)	(3.68)
$\beta_{2,T}$	−0.3505	−0.2039	−0.1632
(t-stat)	(−5.29)	(−7.26)	(−6.17)
R^2	0.1320	0.1058	0.0790

Economic message. The volatility-basis relation is not monotone. Volatility rises at both tails of the forward-curve slope distribution.

5.3 Figure 3: Nonparametric V-shape

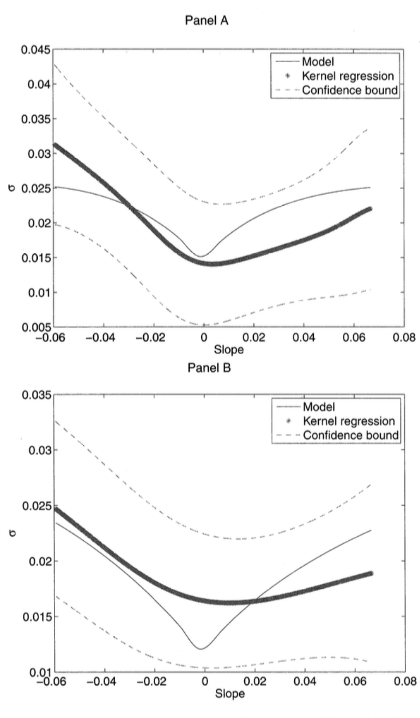

Figure 3. V-shape of volatility of futures prices, data and model output. Futures volatility is plotted as a function of the slope of the forward curve for different values of time to maturity T. Time to maturity is measured in months. In Panel A, $T = 2$; in Panel B, $T = 4$; in Panel C, $T = 6$; in Panel D, $T = 8$. We use Receptive Field Weighted Regression (RFWR), based on Atkeson, Moore, and Schaal (1997) to construct the functional form of volatility implied by the data. In both cases, data and model, volatility exhibits a V-shape pattern as a function of the slope. The slope is demeaned. Volatility is expressed in annual terms. Model parameters are given in Table III.

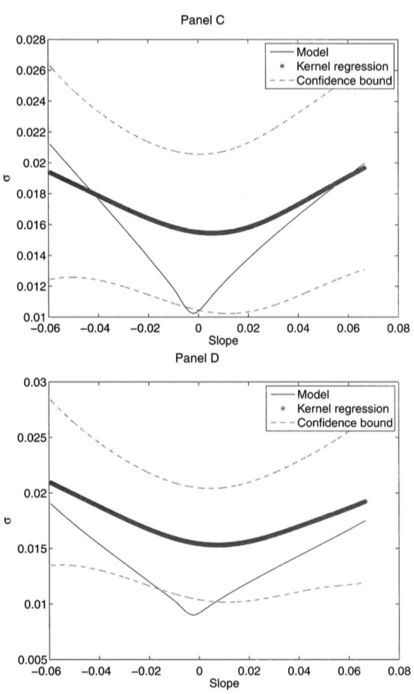

Figure 3. Continued

Read:

- The figure plots nonparametric estimates of volatility as a function of the demeaned slope.
- It reports results for maturities of 2, 4, 6, and 8 months.
- Each maturity shows a clear V-shaped relation.
- The model-generated curve is plotted alongside the empirical curve.

Economic message. The V-shape is not an artifact of a piecewise-linear functional form. It is a visible feature of the data.

5.4 Table II: GARCH robustness

Read:

- The authors control for fitted GARCH(1,1) volatility.
- The GARCH volatility coefficient is large and positive, as expected.
- The V-shape coefficients remain significant: for example, at 1 month they are 0.1610 and −0.1605 in the fitted-GARCH specification.
- In the rolling-GARCH prediction specification, the V-shape coefficients remain positive on the positive side and negative on the negative side.

Economic message. The forward-curve slope contains information about volatility beyond standard volatility persistence.

Table II

GARCH Regressions, Crude Oil Futures

This table reports results for two different regressions. The data are daily returns percentage price changes on NYMEX crude oil (CL) futures from 1985 to 2000. The volatility time series $\sigma_{GARCH}(t, T)$ is obtained by fitting a GARCH(1,1) model to the daily percentage price changes of maturity T. The volatility time series $\widehat{\sigma}_{GARCH}$ is obtained by fitting a GARCH(1,1) model to the time series of daily returns percentage price changes of maturity T up to day $t-1$ and then using it to predict the volatility on day t, $\widehat{\sigma}_{GARCH}(t, T)$. The initial window is set to 300 days. All other variables are defined in the caption to Figure 2.

	1 Month	5 Months	10 Months
$\lvert R(t,T)\rvert = \alpha_T + \beta_{G,T}\sigma_{GARCH}(t,T) + \beta_{1,T}(\widetilde{SL}(t-1))^+ + \beta_{2,T}(\widetilde{SL}(t-1))^- + \varepsilon_T(t)$			
$\beta_{G,T}$	0.5323	0.6145	0.6612
(t-stat)	(8.85)	(9.48)	(12.47)
$\beta_{1,T}$	0.1610	0.0823	0.0595
(t-stat)	(4.32)	(4.01)	(3.81)
$\beta_{2,T}$	−0.1605	−0.0796	−0.0599
(t-stat)	(−2.84)	(−2.92)	(−2.91)
R^2	0.2601	0.2605	0.2495
$\widehat{\sigma}_{GARCH}(t,T) = \alpha_T + \beta_{G,T}\widehat{\sigma}_{GARCH}(t-1,T) + \beta_{1,T}(\widetilde{SL}(t-1))^+ + \beta_{2,T}(\widetilde{SL}(t-1))^- + \varepsilon_T(t)$			
$\beta_{G,T}$	0.9403	0.9748	0.9739
(t-stat)	(68.82)	(155.06)	(161.50)
$\beta_{1,T}$	0.0269	0.0079	0.0064
(t-stat)	(3.06)	(3.34)	(2.90)
$\beta_{2,T}$	−0.0228	−0.0060	−0.0053
(t-stat)	(−3.04)	(−2.23)	(−1.83)
R^2	0.9108	0.9627	0.9586

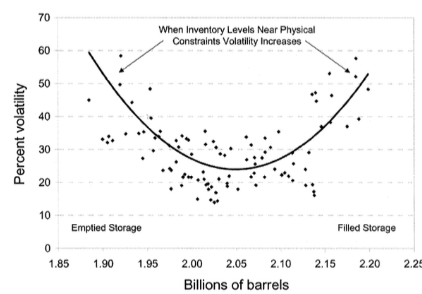

Figure 4. Volatility and supply capacity constraints. The figure gives actual WTI crude oil futures volatility versus expected end-of-March Atlantic Basin inventories. Source: Strongin, Currie, and Fleischmann (1999).

5.5 Figure 4: Capacity constraints and volatility

Read:

- The figure, taken from industry evidence, shows a V-shaped relation between oil futures volatility and expected end-of-March Atlantic Basin inventories.

- Volatility rises when inventories are very low and also when inventories are very high.
- The paper interprets this as related to capacity constraints and supply flexibility.

Economic message. Extreme physical-market states are high-volatility states because adjustment is difficult at both ends.

5.6 Figure 5: Futures prices from the model

Read:

- The figure plots model-implied futures curves for different spot-price states.
- If the spot price is below the investment trigger, the curve reflects expected recovery as low investment and depreciation reduce excess capacity.
- If the spot price is above the trigger, the curve reflects expected mean reversion as maximum investment expands capacity.
- Longer-maturity futures prices are less sensitive to current spot-price deviations.

Economic message. The slope of the forward curve summarizes how far the current spot price is from the model's long-run investment-trigger region.

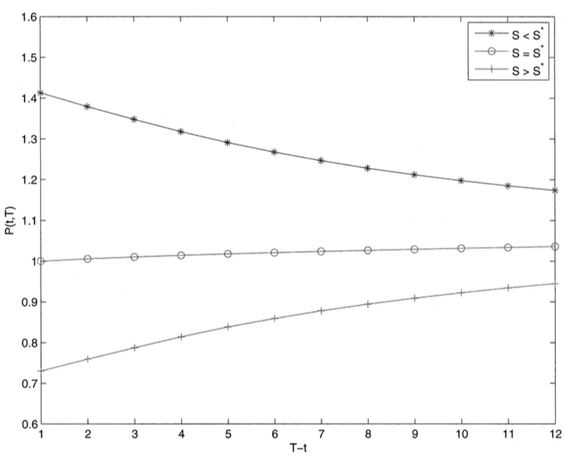

Figure 5. Futures prices, model output. Futures prices are generated by the model using parameter values reported in Table III. The investment threshold, S^*, is normalized to one.

Table III

Parameter Estimates, Crude Oil Futures

This table reports our parameter values. We use a two-step SMM procedure to estimate a vector of seven structural parameters $\theta \equiv \{\gamma, \mu_Y, \sigma_Y, \bar{i}, r, \delta, \lambda\}$. Since only five model parameters can be independently identified from the data (see Section III.A.2), we fix δ and r and estimate the remaining five parameters. We match the unconditional properties of crude oil futures prices, specifically, the historic daily return on fully collateralized 3-month futures position; the unconditional volatility of daily percent price changes for futures of various maturities; as well as the mean, volatility, and 30-day autoregressive coefficient of the slope of the forward curve. In addition, we match the expected growth rate of crude oil consumption. The standard errors are reported where applicable.

Parameter	Value	Std. Error
γ	3.1484	0.2396
$\bar{i}$	0.2362	0.0531
r	0.0200	NA
μ_Y	0.01438	0.0048
σ_Y	0.1043	0.0216
δ	0.1200	NA
λ	2.47×10^{-4}	1.4×10^{-4}

5.7 Table III: Parameter estimates

Read:

- $\gamma = 3.1484$ with standard error 0.2396.
- $\bar{\imath} = 0.2362$ with standard error 0.0531.
- $r = 0.0200$ is fixed.
- $\mu_y = 0.01438$ with standard error 0.0048.

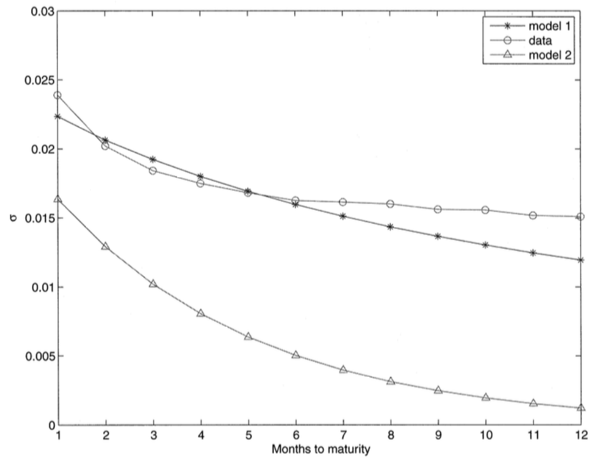

Figure 6. Unconditional volatility of futures prices, model output. The unconditional standard deviation of daily percentage changes in futures prices is constructed based on the output from the model (model 1). The model is fitted to the data on crude oil futures (CL) using a two-step simulated method of moments described in Section III.A.1. Parameter values are reported in Table III. Model 2 corresponds to the unconditional standard deviation of percentage changes in futures prices implied by equation (23). We report averages across 2,000 simulations.

- $\sigma_y = 0.1043$ with standard error 0.0216.
- $\delta = 0.1200$ is fixed.
- $\lambda = 2.47 \times 10^{-4}$ with standard error 1.4×10^{-4}.

Economic message. The estimated parameters imply inelastic demand and bounded capacity adjustment, which are exactly the ingredients needed to generate state-dependent volatility.

5.8 Figure 6: Unconditional volatility, data and model

Read:

- The figure compares the term structure of unconditional futures-price volatility in the data and the model.
- The model captures the slow decline of volatility with maturity.
- A simple reduced-form mean-reverting spot-price model generates volatility that decays too quickly.

Economic message. Production constraints help explain why longer-maturity futures remain volatile.

5.9 Table IV and Figure 7: Conditional volatility from the model

Read:

- The model reproduces negative coefficients in the misspecified linear volatility-basis regression.
- More importantly, it reproduces the opposite signs in the piecewise V-shape regression.

Table IV

Conditional Volatility, Model Output

Conditional variance of crude oil futures prices is compared with the output of the model under two different econometric specifications. See the caption to Figure 2 for a detailed description of the underlying variables.

	1 Month		5 Months		10 Months	
	Data	Model	Data	Model	Data	Model
	$\lvert R(t,T)\rvert = \alpha_T + \beta_T \widetilde{SL}(t-1) + \varepsilon_T(t)$					
β_T	−0.0743	−0.0030	−0.0462	−0.0100	−0.0419	−0.012
	$\lvert R(t,T)\rvert = \alpha_T + \beta_{1,T}(\widetilde{SL}(t-1))^+ + \beta_{2,T}(\widetilde{SL}(t-1))^- + \varepsilon_T(t)$					
$\beta_{1,T}$	0.3191	0.0854	0.1791	0.1482	0.1308	0.1246
$\beta_{2,T}$	−0.3505	−0.0964	−0.2039	−0.1822	−0.1632	−0.1651

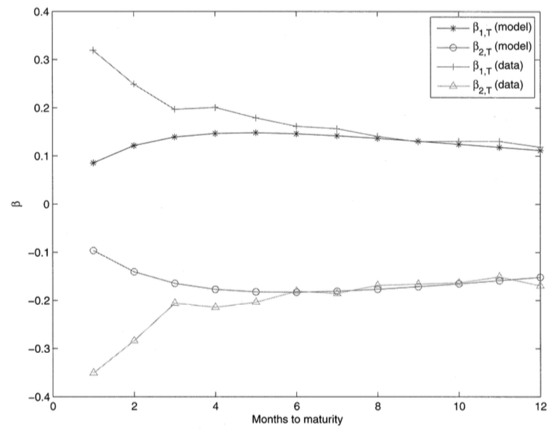

Figure 7. Conditional volatility of futures prices, model output. For each time to maturity, T, we simulate a time series of daily futures prices using parameter values reported in Table III. We then compute daily percentage changes in futures prices, denoted by $R(t, T)$. The instantaneous volatility of futures prices is related to the beginning-of-period slope of the forward curve, defined as

$$SL(t) = \ln\left[\frac{P(t-1,6)}{P(t-1,3)}\right],$$

Table V

Sensitivity of Model Moments to Parameters

This table presents elasticities of model moments with respect to the model parameters. The baseline parameters are given in Table III. Each elasticity is calculated by simulating the model twice: once with a value of the parameter of interest 10% of one standard deviation below its baseline value, and once with a value 19% of one standard deviation above its baseline value. Then the change in the moment is calculated as the difference between the results from the two simulations. This difference is then divided by the change in the underlying structural parameter between the two simulations. The result is then multiplied by the absolute value of the ratio of the baseline structural parameter to the baseline moment.

	Data	Baseline Moments	γ	$\bar{\imath}$	μ_Y	σ_Y	$\lambda(\times 10^{-2})$
Average Slope of the Forward Curve	−0.0124	0.0065	1.8401	−1.2862	0.1814	2.3471	−0.5004
Volatility of the Slope of the Forward Curve	0.0287	0.0285	0.9704	0.5516	−0.0377	0.7005	0.4189
30-days AR(1) Coefficient of Forward Curve Slope	0.7653	0.8337	−0.0041	−0.1221	−0.0038	0.2112	0.1177
Long Term Crude Oil Consumption Growth	0.0092	0.0092	0.0001	−0.0003	1.5652	−1.1843	2.6939
Unconditional Variance 3-month CL Futures	0.0184	0.0186	0.7444	−0.5535	0.0036	1.2367	0.0646
Unconditional Variance 6-month CL Futures	0.0163	0.0149	0.6807	−0.9276	0.0104	1.4487	0.0625
Unconditional Variance 9-month CL Futures	0.0156	0.0124	0.6176	−1.2162	0.0173	1.5690	0.0706
$\beta_{1,T}$ 3-month CL Futures	0.1968	0.1393	−0.0367	1.2496	−0.1222	−0.1146	−0.0488
$\beta_{1,T}$ 6-month CL Futures	0.1616	0.1462	−0.0634	1.0368	−0.1475	0.4632	−0.1799
$\beta_{1,T}$ 9-month CL Futures	0.1298	0.1309	−0.0795	0.8254	−0.1634	0.9137	−0.0662
$\beta_{2,T}$ 3-month CL Futures	−0.2059	−0.1646	0.0745	−1.7212	0.1753	−0.0084	0.3828
$\beta_{2,T}$ 6-month CL Futures	−0.1808	−0.1829	0.0897	−2.6308	0.2155	0.5637	0.4722
$\beta_{2,T}$ 9-month CL Futures	−0.1659	−0.1713	0.0983	−3.5200	0.2551	1.0333	0.5976

- At 10 months, the data have $\beta_{1,T} = 0.1308$ and $\beta_{2,T} = -0.1632$; the model has 0.1246 and -0.1651.
- The match is especially good for medium and longer maturities.
- The fit is weaker for the shortest maturities, where storage likely matters more.

Economic message. The model's main mechanism is quantitatively relevant, not just qualitatively suggestive.

5.10 Table V: Sensitivity of model moments

Read:

- The table reports elasticities of model moments with respect to structural parameters.
- Increasing σ_y has a large positive effect on volatility moments.
- Increasing γ also raises futures-price volatility.
- Increasing $\bar{\imath}$ tends to reduce futures-price volatility because capacity can adjust more quickly.
- Drift and risk-premium parameters are more important for growth and slope moments than for the volatility level.

Economic message. The model's key comparative statics are intuitive: more demand volatility and less elastic demand raise price volatility, while faster capacity adjustment dampens it.

6 Short summary

- The paper documents a new empirical fact: oil futures volatility is V-shaped in the slope of the forward curve.
- Volatility is high in both steep backwardation and steep contango.
- Standard storage models are not well suited to explain this nonmonotone relation.
- The paper builds a competitive production model with irreversible investment and a maximum investment rate.
- These constraints make supply elasticity time-varying.
- When capacity is far from demand, supply is inelastic and price volatility is high.
- The forward-curve slope is large in absolute value precisely when the current state is far from the investment trigger.
- The model is estimated by simulated method of moments using futures-price and consumption-growth moments.
- The estimated model reproduces the slow decline of futures volatility across maturities and the V-shaped conditional volatility relation.

7 Conclusion

7.1 Core conclusion

Kogan, Livdan, and Yaron show that oil futures volatility is not only related to whether the market is backwardated or in contango. The deeper fact is that volatility is high when the forward-curve slope is large in absolute value.

7.2 Economic intuition

A large absolute slope indicates that the current spot price is far from its long-run level. In those states, one of the investment constraints binds: firms either cannot disinvest when capacity is excessive or cannot expand capacity fast enough when capacity is scarce. Demand shocks are therefore absorbed mainly by prices rather than by quantities.

7.3 Takeaway

The paper's main lesson is that commodity futures pricing should include the production side of the economy. Storage matters, especially at short horizons, but investment constraints and time-varying supply elasticity are essential for understanding the conditional volatility of oil futures prices.